Proceedings of the Combustion Institute 40 (2024) 105609

Contents lists available at ScienceDirect

Proceedings of the Combustion Institute

journal homepage: www.elsevier.com/locate/proci

Identification of entropy waves in a partially premixed combustor

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ARTICLE INFO

Keywords:
Thermoacoustic instability
Entropy waves
Entropy transfer function
Large eddy simulation
System identification

ABSTRACT

Unsteady combustion generates not only acoustic waves, but also fluctuations of the burnt gas temperature — referred to as entropy waves. These waves are convected by the mean flow through the combustor and result in conversion to acoustic energy when accelerated in an exit nozzle. The upstream traveling acoustic wave can then couple with the unsteady heat release of the flame and cause self-excited thermoacoustic instability, particularly at low frequencies (“rumble”). In this work, large eddy simulation (LES) is combined with system identification (SI) to determine the entropy transfer function (ETF) of a partially premixed, swirl-stabilized flame with hydrogen enrichment. We compare the single-input single-output (SISO) entropy transfer function identified from a broadband-forced LES with air mass flow modulation to the one obtained experimentally through tunable diode laser absorption spectroscopy with wavelength modulation spectroscopy (TDLAS-WMS) to measure temperature fluctuations. Then, multiple-input single-output (MISO) identification is applied to time series data obtained from simultaneous modulation of air and fuel mass flow to estimate the individual contributions of perturbations in velocity and equivalence ratio to entropy response. Equivalence ratio fluctuations are found to be the dominant generation mechanism of entropy waves. Finally, the entropy transfer function is identified at various positions in the combustion chamber to analyze the decay of entropy waves governed by convective dispersion.

1. Introduction

Unsteady combustion may generate coherent temperature fluctuations in the burnt gas [1], also known as *entropy waves*. These inhomogeneities are convected through the combustion chamber and attenuated by turbulent mixing and spatial variations in the velocity field [2,3]. In aero-engine combustors, where convective dispersion is slow relative to the residence time, entropy waves often reach the turbine stage with high amplitudes, where they are accelerated [4–6]. This acceleration results in a mode conversion into pressure oscillations [7]. On the one hand, the downstream traveling acoustic waves add to the overall noise emissions of an aircraft as indirect combustion noise [6]. On the other hand, the upstream propagating acoustic waves might interact constructively with the unsteady combustion and cause thermoacoustic combustion instability, particularly at low frequencies (“rumble”) [1,8]. This coupling is achieved through multiple connected mechanisms involving fluctuations of entropy, pressure, velocity and fuel concentration [8].

In this work, lean *partially premixed* (or technically premixed) turbulent flames are considered, where the fuel is injected close to the burner outlet resulting in spatial mixture inhomogeneities at the flame. Equivalence ratio fluctuations can contribute to combustion instability only if driven in a coherent manner by the combustion process and pressure oscillations, thus closing a positive feedback loop [9,10]. Coherent fluctuations in fuel concentration originate from pressure and velocity fluctuations at the location of the fuel injection [11], which are convected to the flame [2]. These locally richer and leaner pockets of fresh mixture cause local changes in the adiabatic flame temperature, i.e. entropy waves [8]. In the absence of equivalence ratio fluctuations, entropy waves may occur due to unsteady wall heat loss (dominant effect) or differential diffusion, as shown in our previous study on entropy waves in a fully premixed non-adiabatic combustor with hydrogen enrichment [12]. These mechanisms are coherent with velocity fluctuations.

The two excitation mechanisms mentioned above and their interaction result in a frequency-dependent response of entropy to flow perturbations, known as entropy transfer function (ETF). The model of a partially premixed combustion system with an acoustically stiff fuel injection (large pressure drop across fuel injectors) located in acoustically compact distance (much shorter than the acoustic wavelength) to the burner outlet reduces to a single-input single-output (SISO) model structure [11,13] (see Fig. 1(a)). This black-box model implicitly accounts for fluctuations in equivalence ratio caused by

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https://doi.org/10.1016/j.proci.2024.105609
Received 4 December 2023; Accepted 2 July 2024
Available online 2 August 2024

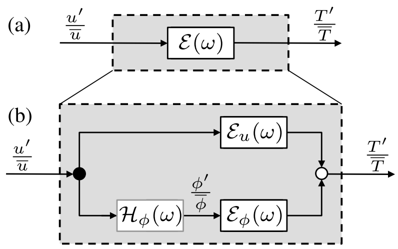

Fig. 1. Black-box model structure of (a) overall entropy transfer function $\mathcal{E}(\omega)$ and (b) individual contributions $\mathcal{E}_\phi(\omega)$ and $\mathcal{E}_u(\omega)$ in a partially premixed combustor with stiff and acoustically compact fuel injection. $\mathcal{H}_\phi(\omega)$ relates fluctuations in equivalence ratio ϕ' and velocity u'.

velocity perturbations at the fuel injector. For the configuration considered here, the forcing of air mass flow determines the overall ETF in partially premixed flames [14]. In this work, high-fidelity large eddy simulation (LES) is combined with system identification (SI) [15] to determine the frequency response of entropy waves. The simulations are complemented by tunable diode laser absorption spectroscopy with wavelength modulation spectroscopy (TDLAS-WMS) [14] to measure the overall ETF.

This study aims at enhancing the understanding of entropy response in partially premixed flames by separating the contributions of equivalence ratio and velocity fluctuations, respectively, as shown in Fig. 1(b). Therefore, a multiple-input single-output (MISO) structure is introduced for the ETF, similar to that proposed by Huber and Polifke [11] for the flame transfer function (FTF), which describes the response of heat release rate to perturbation in velocity and equivalence ratio. The data for the MISO approach is generated by a LES with simultaneous excitation of the air and fuel mass flow with uncorrelated broadband signals. The next section describes the experimental and numerical setup of the combustor used in this work. Section 3 provides information on the methodology for SISO and MISO model identification. Section 4 presents the ETF obtained by LES/SI and experiments and analyzes the individual transfer functions. Finally, the decay of entropy waves along the combustor is investigated.

2. Experimental and numerical setup

2.1. The ETH combustor

To investigate the entropy response of partially premixed flames to modulations in air mass flow, experimental results are compared to acoustically forced large eddy simulation. In this section, a brief overview of the laboratory-scale test rig [14,16] of a turbulent swirl combustor shown in Fig. 2(a) is provided. The test rig has a modular layout and consists of a square air plenum and combustion chamber (62×62 mm^2), an annular burner (100 mm length) equipped with an axial swirler, and a motor-driven adjustable outlet orifice to ensure thermoacoustic stable conditions. Some of these modules consist of air-cooled quartz windows for optical access, the rest of water-cooled aluminum walls.

The air is supplied at ambient temperature and pressure through the plenum and subsequently passes through eight axial swirl vanes. The methane-hydrogen fuel mixture is injected in two stages at approximately choked conditions (unity Mach number) as jet-in-cross-flow[1] into the turbulent swirling air flow within the burner annulus downstream of the swirler. Both stages consist of four small injection holes (0.8 mm diameter), positioned at 90 degree increments around the center body. A large pressure drop across the injectors ensures acoustically stiff fuel injection. The short distance (50 mm and 22 mm) between the injector stages and the burner outlet (also known as burner mouth) leads to spatial and temporal inhomogeneities of equivalence ratio before combustion.

The air mass flow is mono-frequently excited through an upstream loudspeaker. Six G.R.A.S. microphones are located between loudspeaker and burner to reconstruct the acoustic field. Two different lasers centered at around 1392 nm and 1469 nm that targets the H_2O absorption spectra are employed for the TDLAS-WMS method. The beams are combined together with a fiber combiner. Optical access for the laser beam is located 280 mm downstream of the burner outlet as shown in Fig. 2(a). The laser beam passes the hot gas section and is detected by a photodiode. The raw signals from the photodiode are converted to temperature signals. The ETF is then obtained by dividing these signals by the reconstructed acoustic velocity. A more detailed description of the TDLAS-WMS setup and algorithm are presented in [14].

The methane-hydrogen-fired combustor (59% hydrogen by volume) was operated at an equivalence ratio of $\phi = 0.78$ and an air mass flow of 18.2 g/s resulting in a thermal power of 43.7 kW. The operating condition considered in this work corresponds to "OP 3" in [16] and provides additional measurement data including line-of-sight integrated OH^*chemiluminescence and planar laser-induced fluorescence of the OH radical (OH-PLIF). The flame images are compared with the numerical results in Section 4.1.

2.2. LES setup

Large eddy simulations of the ETH combustor shown in Fig. 2 are performed using the massively parallel explicit cell-vertex AVBP code (www.cerfacs.fr/avbp7x/index.php), which solves the multi-species compressible Navier–Stokes equations. Subgrid Reynolds stresses are modeled using the SIGMA turbulent closure [17] without wall functions. A second-order (in both space and time) Lax–Wendroff scheme is used to discretize the convective terms, found to be sufficient for LES/SI applications [15]. The time step is fixed to $\Delta t = 1.3 \times 10^{-8}$ s, assuring an acoustic Courant-Friedrichs-Lewy (CFL) number below 0.8 (corresponds to a hydrodynamic CFL number below approx. 0.4). The computational domain shown in Fig. 2(b) consists of the burner (axial swirler, fuel injectors and mixing duct) and 300 mm of the combustion chamber and is spatially discretized with 72.8 million tetrahedral elements. The grid is shown in Fig. 2(c) and refined as follows: Δx = 80 μm in the injectors and fuel jets, Δx = 250 μm in the swirler and flame region (laminar flame thickness is about 397 μm), Δx = 300 μm in the mixing duct, and Δx = 400 μm at the combustor walls.

Inlet and outlet boundaries are modeled with the Navier–Stokes characteristic boundary conditions (NSCBC) [18] imposing mass flow rates (see numbers in Section 2.1) and ambient pressure, respectively, with relaxation coefficients $K_{\text{air}} = 5.0 \times 10^4$ s^{-1}, $K_{\text{fuel}} = 5.0 \times 10^5$ s^{-1} and $K_{\text{outlet}} = 5.0 \times 10^2$ s^{-1}. The inlet and outlet impedances are set to ensure thermoacoustic stability and do not have to correspond to their actual values in the experiment. To suppress a self-excited transverse chamber mode, a characteristic zero mass-flux boundary condition with relaxation towards a target temperature is imposed at the perimeter of the backplane as successfully applied by Eder et al. [19]. Heat loss $\dot{q} = \left(T_{\text{wall}} - T_\infty\right)/R_{\text{w}}$ is imposed at the remaining backplane, combustion chamber side walls and the center body tip, where the thermal resistance R_{w} is adjusted to ensure wall temperatures around $T_{\text{wall}} = 850$ K.

The analytically reduced chemistry (ARC) mechanism presented and validated by Laera et al. [20] for CH_4-H_2-air combustion is employed. It consists of 20 transported species, 166 reactions, and 9 quasi-steady-state species derived from the detailed GRI-Mech 3.0. Supplementary material in [20] offers further information on the ARC mechanism.

[1] The momentum flux ratio is estimated based on LES results as $J = \left(\rho_{\text{fuel}} u_{\text{fuel}}^2\right) / \left(\rho_{\text{air}} u_{\text{air}}^2\right) \approx 100$.

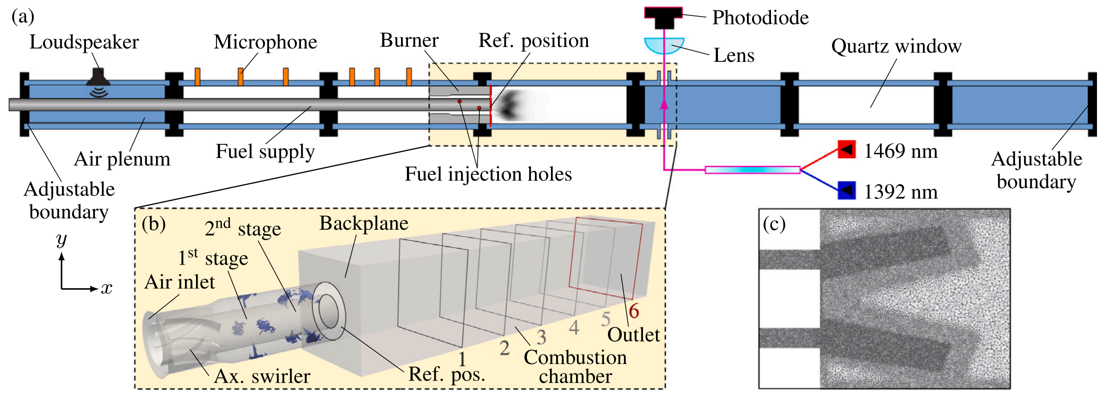

Fig. 2. (a) Schematic of the swirl combustor equipped with TDLAS-WMS and microphone arrays for ETF measurement. The reference position for velocity reconstruction is indicated at the burner outlet. Stiff fuel injection of the CH_4-H_2 mixture into the turbulent swirling air flow is done inside the burner (partially premixing). (b) Computational domain used in LES with instantaneous iso-contours of H_2 to visualize the staged fuel injection together with the reference positions for velocity and temperature reconstruction (position 6 indicates location of ETF measurement). (c) Grid refinement in the flame region.

Transport properties are described by a constant mixture Prandtl number and constant Schmidt numbers for each species [20] to account for differential diffusion. Turbulence-combustion interaction is modeled with the dynamic thickened flame (DTFLES) model [21], where the flame is thickened based on a flame sensor (5 points in the flame are specified as target resolution). The model parameters (laminar flame speed, laminar flame thickness, reaction rates) are functions of the equivalence ratio.

3. Methodology

3.1. Entropy disturbances

Based on the linearized Gibbs equation and neglecting compositional inhomogeneities [22], together with the linearized expression for the ideal gas law $\rho'\overline{\rho} = p'/\overline{p} - T'/\overline{T}$, entropy fluctuations s' can be approximated by fluctuations of the temperature T' and pressure p',

$$\frac{s'}{c_p} = \frac{1}{\gamma}\frac{p'}{\overline{p}} - \frac{\rho'}{\overline{\rho}} \approx \frac{T'}{\overline{T}} - \frac{\gamma-1}{\gamma}\frac{p'}{\overline{p}}\ , \tag{1}$$

where ρ, c_p and γ are the density, the specific isobaric heat capacity and the heat capacity ratio, respectively, and $(\cdot)'$ are small fluctuations around its temporal mean value $\overline{(\cdot)}$. Assuming low Mach numbers and no large amplitude acoustic fluctuations, the contribution of pressure fluctuations (second term on the right-hand side of Eq. (1)) is known to be small in premixed flames. This simplifies the right-hand side of Eq. (1) to [6,8,14]

$$\frac{s'}{c_p} \approx \frac{T'}{\overline{T}}\ . \tag{2}$$

The time series of the temperature fluctuations are inferred in the experiment using TDLAS-WMS at a reference position (0.28 m from the burner outlet). In the LES, they are recorded as an area-integral at several planes along the combustion chamber (and also at the measurement position). The reference position in the experiment was chosen to be sufficiently far from the reaction zone, while still ensuring that the temperature variations were detectable. In technically relevant applications, entropy waves are only problematic if they have sufficient amplitudes at the combustor exit/turbine inlet to make a relevant contribution to the overall engine noise or trigger a self-excited feedback loop.

3.2. Entropy transfer function

The frequency-dependent response of coherent fluctuations of the burnt gas temperature to coherent velocity fluctuations can be related for small amplitudes using an entropy transfer function,

$$\frac{T'}{\overline{T}} \equiv \mathcal{E}(\omega)\frac{u'_{\mathrm{ref}}}{\overline{u}_{\mathrm{ref}}}\ , \tag{3}$$

where u_{ref} is the velocity at a reference location (burner outlet), $\mathcal{E}(\omega)$ is the entropy transfer function, and $\omega = 2\pi f$ is the Fourier transform variable (angular frequency). The transfer function in Eq. (3) can be understood as the overall entropy response, which contains both the mechanisms of equivalence ratio and velocity fluctuations. In partially premixed combustion systems with stiff fuel injection (constant fuel mass flow), velocity fluctuations at the injector (caused by modulation of the air mass flow) lead to phase-opposite equivalence ratio fluctuations at the injector [8,23,24] that are convected in the mixing duct toward the flame [2]. If, in addition to the stiff fuel injection, an acoustically compact mixing duct is present (both are given in the present configuration), a single-input single-output model structure for the entropy response is justified [11,13]. In both the experiment and a first LES, the overall ETF in Eq. (3) is determined by an upstream modulation of the air mass flow via mono-frequency (experiment) and broadband (LES) forcing. The experimental and numerical results are compared in Section 4.1.

As proposed by Huber and Polifke [11] for the flame transfer function, a multiple-input single-output model structure is utilized in the present work for the ETF,

$$\frac{T'}{\overline{T}} \equiv \mathcal{E}_\phi(\omega)\frac{\phi'_{\mathrm{ref}}}{\overline{\phi}_{\mathrm{ref}}} + \mathcal{E}_u(\omega)\frac{u'_{\mathrm{ref}}}{\overline{u}_{\mathrm{ref}}}\ , \tag{4}$$

where $\mathcal{E}_\phi$ (independent of impedance and position of the fuel injector) and $\mathcal{E}_u$ represent the individual entropy responses to equivalence ratio and velocity fluctuations at the burner outlet (reference position). In Eq. (4) all mechanisms generating entropy waves by global equivalence ratio fluctuations at the burner outlet are represented by $\mathcal{E}_\phi$. All generation mechanisms of entropy waves present in the system due to perturbations of the incoming flow at the flame base (e.g. wall heat losses [12,25,26], differential diffusion [12], dilution jets [25] or incomplete combustion [26]) are represented by $\mathcal{E}_u$. Separating the input quantities permits independent consideration of both excitation mechanisms. The present experimental setup is not equipped for excitation in the fuel supply duct and therefore is not suitable for the MISO analysis. However, simultaneous broadband forcing of air and fuel mass flow is applied in a second LES. Uncorrelated excitation signals for both channels (air and fuel mass flow) allow to distinguish the individual contributions using MISO identification [11,27] even if the eventual input signals (u'_{ref}, ϕ'_{ref}) exhibit a certain correlation [28]. The reader may refer to [11,28] for further information on the MISO identification. The individual contributions $\mathcal{E}_\phi$ and $\mathcal{E}_u$ are identified from LES data in Section 4.2.

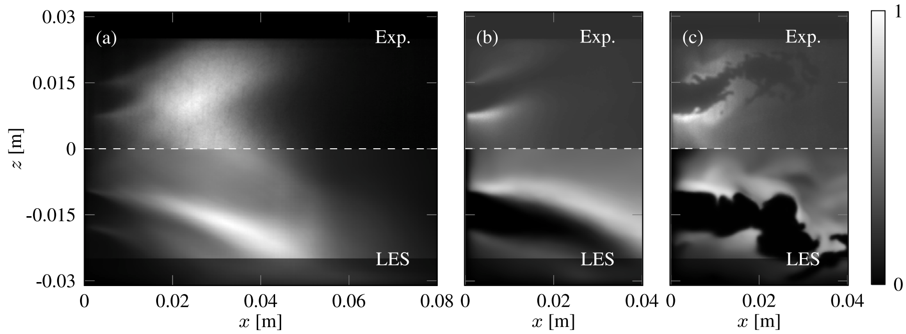

Fig. 3. Comparison of flame shapes between experiment [16] (top) and LES (bottom), with (a) line-of-sight (LOS) integration of time-averaged OH* chemiluminescence image (Exp.) and HRR (LES), (b) time-averaged OH-PLIF signal (exp.) and OH mass fraction $\overline{Y}_{OH}$ (LES), and (c) instantaneous OH-PLIF (exp.) and OH mass fraction Y_{OH} (LES).

Under the same assumptions as in Eq. (3) (stiff and acoustically compact fuel injection), the overall ETF is related to the two contributions $\mathcal{E}_\phi$ and $\mathcal{E}_u$ via [29]

$$\frac{T'}{\overline{T}} \equiv \underbrace{\left[\mathcal{H}_\phi(\omega)\mathcal{E}_\phi(\omega) + \mathcal{E}_u(\omega)\right]}_{\mathcal{E}(\omega)} \frac{u'_{\mathrm{ref}}}{\overline{u}_{\mathrm{ref}}} \,. \tag{5}$$

The transfer function $\mathcal{H}_\phi(\omega)$ represents the equivalence ratio response to velocity fluctuations at the burner outlet and can be obtained with system identification from the LES with only air mass flow forcing (ϕ'_{ref} and u'_{ref} are correlated) via

$$\frac{\phi'_{\mathrm{ref}}}{\overline{\phi}_{\mathrm{ref}}} \equiv \mathcal{H}_\phi(\omega)\frac{u'_{\mathrm{ref}}}{\overline{u}_{\mathrm{ref}}} \,. \tag{6}$$

Eq. (5) is used in Section 4.3 to validate the MISO model (Eq. (4)) with experimental data.

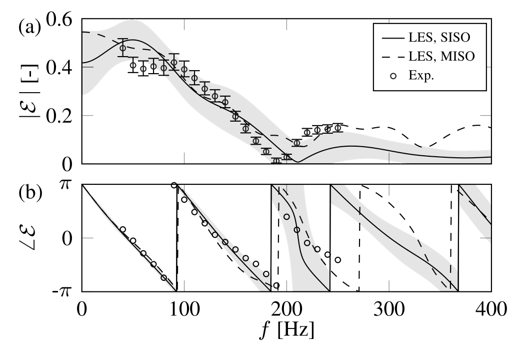

Fig. 4. (a) Gain and (b) phase of overall entropy transfer functions obtained from experiment, and SISO and MISO identification. The error bars and shaded areas represent 95% confidence intervals.

4. Results and discussion

4.1. Comparison of flame shapes

A comparison of time-averaged and instantaneous flame images from LES and optical measurements, respectively, is shown in Fig. 3. The mean fields in LES are time-averaged over a duration of 35 ms. Although OH* light intensity is not a quantitative indicator for heat release rate in partially premixed flames (non-linear response of OH* to ϕ') [30], it may serve as a qualitative measure to compare the shape and characteristics of the flame from experiment and LES. As shown in Fig. 3(a), the high volume fraction of hydrogen in the fuel ($X_{H_2} = 0.59$) causes a short flame with dominant M-shape that exhibits pronounced reactions in the inner and outer shear layer, which is well captured in LES. Furthermore, time-averaged and instantaneous OH-PLIF images (experiment) are compared with OH mass fractions (LES), see Fig. 3(b) and (c), respectively. Both flames show an accumulation of OH in the inner shear layer, more pronounced in the LES. As noticeable in Fig. 3(a) and (b), the flame length is clearly overestimated compared to experiments, which may be partially attributed to the simplifications of the thermal boundary modeling and associated inaccuracies in the absolute wall temperature and heat distribution. Further. Nevertheless, the agreement between experiment and LES was found to be acceptable for the subsequent entropy response analysis.

4.2. Overall entropy response

In the first step, the air mass flow is excited in experiment and LES to determine from the temperature signal at the measurement position (0.28 m from the burner outlet) the overall entropy transfer function as introduced in Eq. (3). The frequency response $\mathcal{E}(\omega)$ is typically shown in gain and phase representation (Bode plot), $\mathcal{E}(\omega) = |\mathcal{E}(\omega)|e^{i\angle\mathcal{E}(\omega)}$, where $\omega \in \mathbb{R}$ is the (real-valued) angular frequency. Fig. 4 shows the ETF estimated with a finite impulse response (FIR) filter [15,31] from time series data of upstream velocity (burner outlet) and downstream temperature fluctuations. The error bars indicate the standard deviation of $\sigma = 1.96$ (95% confidence interval according to Sovardi et al. [32]) for the measured and predicted transfer functions to account for the aleatoric uncertainty of the spectroscopic database and the coefficients of the system identification, respectively. Since system identification is a data-driven method, the confidence intervals become smaller with increasing data size (length of the time series). The predicted gain of the ETF in Fig. 4(a) shows quantitative agreement with experiment. It exhibits an overall low-pass behavior due to the convective nature of entropy waves and decays towards zero gain after a local maximum around 250 Hz. Possible reasons for the non-monotonically decaying gain are presented in [14] for a similar flame, such as hydrodynamic perturbations generated at the swirler, which might constructively or destructively interact with the equivalence ratio perturbations at the injector locations, or the existence of large coherent structures in swirling flows. As shown in Section 4.4, this characteristic feature vanishes when identifying the ETF at more upstream positions, suggesting that it is related to a mechanism caused by convective dispersion.

The predicted and measured ETF phases are shown in Fig. 4(b). The following presents a simplified argument for the low frequency-limit of a lean flame, where the flame is assumed to be the sole source of entropy generation. The low-frequency limit $\omega \to 0$ can be thought of as a change in one quantity (e.g. $\delta\dot{m}_{\mathrm{air}}$) resulting in a new steady state of the system [33]. Since an increase in air mass flow ($\delta\dot{m}_{\mathrm{air}}$, i.e. $\delta u > 0$) corresponds to an increased amount of air available for combustion relative to the fuel (excess air becomes more pronounced), the burnt gas temperature ($\delta T < 0$) reduces. Therefore u' and T' are out of phase and $\angle\mathcal{E}(\omega \to 0) \to \pm\pi$, which is consistent with results

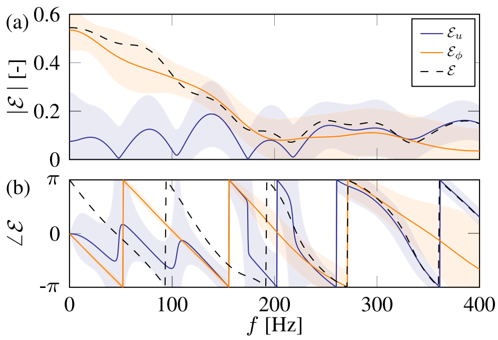

Fig. 5. (a) Gain and (b) phase of individual entropy transfer functions $\mathcal{E}_u$ and $\mathcal{E}_\phi$ from MISO identification. The shaded areas represent 95% confidence intervals. $\mathcal{E}$ is reconstructed via Eq. (5).

from LES/SI and experiment[2]. In addition, the slope of the phase is very well predicted, indicating accurate modeling of the bulk velocity transport in the LES. The phase of the ETF corresponds to the time delay between the acoustic perturbation at the reference position and the temperature fluctuations at the measuring point, which matches well with the expected convective time delay.

4.3. MISO identification of entropy response

In the next step, simultaneous air and fuel mass flow forcing is applied in LES using uncorrelated excitation signals for the two channels [11]. The individual entropy responses to equivalence ratio and velocity fluctuations at the burner outlet are directly obtained from multiple-input single-output (inputs: ϕ'_{ref} and u'_{ref}, output: T') identification using a FIR model structure. The overall number of model parameters (see e.g. [34] for more information on their choice and estimation), i.e. FIR coefficients, is approximately doubled in comparison to the SISO model (Section 4.2). This results in larger confidence intervals for the same time series length [27] (see [32] for more information on the uncertainty propagation), shown in Fig. 5. Eq. (5) is utilized to reconstruct the overall ETF from the individual contributions $\mathcal{E}_\phi$ and $\mathcal{E}_u$. The reconstructed ETF, additionally shown in Fig. 4 (dashed line), exhibits very similar gain and phase as the ETF obtained with SISO identification (solid line). This validates the MISO methodology and the underlying separation of excitation mechanisms. Some of the differences between the estimated entropy responses may be attributed to the comparably larger confidence intervals of the individual responses $\mathcal{E}_\phi$ and $\mathcal{E}_u$ discussed below.

The contribution from fluctuations in equivalence ratio $\mathcal{E}_\phi$ is significantly larger and dominates the overall entropy response $\mathcal{E}$ over almost the entire frequency range, particularly below 200 Hz. This finding supports the arguments found in the literature [1,2,8,26] that equivalence ratio fluctuations are the dominant mechanism for the generation of entropy waves in partially premixed flames. The contribution of entropy response due to velocity fluctuations $\mathcal{E}_u$ (see Fig. 5) exhibits a similar order of magnitude and shape as observed in our previous study on the fully premixed configuration of the same combustor (different operating point) [12]. This is reasonable, as the studied combustor involves severe wall heat losses and considerable differential diffusion due to the high volume fraction of hydrogen.

[2] Note that the phase of the ETF in the original experimental work by Dharmaputra et al. [14] (Figs. 6 and 7) is erroneously shifted by π.

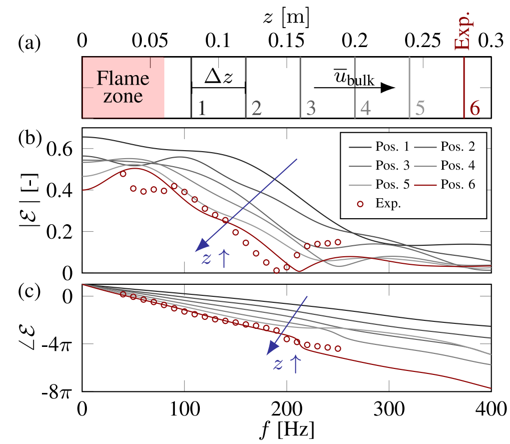

Fig. 6. (a) Sampling positions 1 to 6 in the combustion chamber. Position 6: location of the experimental measurements. (b) Gain and (c) phase of overall entropy transfer functions. Confidence intervals (not shown) are comparable to Fig. 4.

4.4. Convective transport of entropy waves

The decay of entropy waves is just as crucial as their actual excitation mechanisms discussed in the previous two subsections, as entropy waves only affect the thermoacoustic feedback cycle due to mode conversion if they are not fully attenuated during the convection through the combustor. Various relevant mechanisms associated with dispersion and dissipation are discussed in the works of Sattelmayer [2], Morgans et al. [5], Weilenmann et al. [35], Polifke [36], Kaiser and Oberleithner [37,38] and Kaiser et al. [39], whereas each burner type exhibits different behavior due to its complex flows (e.g. swirl, flow separation, central vortex core) and unique geometry. This repeatedly leads to contradictory conclusions in the literature on the importance of entropy waves in technically relevant combustors [6,35]. Consequently, the decay of entropy waves in the present turbulent swirl combustor is assessed in the following.

We record coherent temperature fluctuations to upstream air mass flow modulations at five additional monitoring planes (see positions 1 to 6 in Fig. 2(b) or Fig. 6(a)) between the reaction zone and the measurement position to identify the overall entropy responses via Eq. (3) at these locations, shown in Fig. 6. The gain exhibits a continuous decrease with increasing residence time in the combustor over the entire frequency range. Nevertheless, for low frequencies (below 100 Hz), the gain is less affected by the attenuation in the combustion chamber and remains similar to regions close to the reaction zone. The phase in Fig. 6(b) has been unwrapped to visualize the convective nature of entropy waves. The phase change along the combustor matches the predicted convective transport, which is constant for a constant bulk velocity $\bar{u}_{\mathrm{bulk}}$ and equal delays $\Delta\tau$ between each of the six positions.

It is beyond the scope of this work to rank the strength of the entropy waves found at the combustor outlet in terms of their relevance to thermoacoustic instability, but Fig. 6 clearly emphasizes that turbulent mixing, shear dispersion and further decay mechanisms in the present combustor are not properly suppressing entropy waves.

5. Conclusion

This work employs both large eddy simulations combined with system identification (LES/SI) and laser-based experiments (TDLAS-WMS) to determine the entropy transfer function of a partially premixed flame. The predicted and measured entropy response shows remarkable agreement in both the gain and phase. Furthermore, the well predicted phase slope suggests a correct modeling of the bulk flow in LES. Subsequently, the individual contributions of entropy response to equivalence ratio and velocity fluctuations are analyzed utilizing multiple-input single-output identification with time series data obtained from simultaneous air and fuel mass flow forcing in LES. The fluctuations of equivalence ratio are found to be the dominant mechanisms for the generation of entropy waves even in the presence of strong water cooling with severe wall heat losses and significant differential diffusion (59% hydrogen by volume) [12], suggesting that this finding is generalizable to other configurations in premixed combustion. Furthermore, this is consistent with literature [1,2,8,26].

In addition, we numerically analyzed the convective transport of entropy waves in the generic partially premixed swirl combustor by evaluating entropy transfer functions on various planes in the combustion chamber. It is demonstrated that the attenuation is weak enough to ensure significant entropy wave strength at the combustor exit. For low-frequencies ($f < 100$ Hz), the gain of the entropy response is comparable to regions shortly downstream of the reaction zone.

To the authors' best knowledge, this is the first time that the dominance of equivalence ratio fluctuations on the generation of entropy waves has been investigated and confirmed *quantitatively* in accordance with [1,2,8,26] using MISO identification, i.e. that the individual excitation mechanisms are studied, and that the entropy transfer function of a partially premixed flame is determined numerically.

Novelty and significance statement

The novelty of this research is the determination of entropy transfer functions in a partially premixed combustor by large eddy simulation combined with system identification. Numerical findings are complemented by experimental data obtained with the Tunable Diode Laser Absorption Spectroscopy with Wavelength Modulation Spectroscopy method (TDLAS-WMS). By further separating the entropy responses to equivalence ratio and velocity fluctuations utilizing multiple-input single-output identification, the dominance of equivalence ratio fluctuations in this process is revealed. Furthermore, the evaluation of entropy transfer functions across various positions within the combustion chamber sheds light on the decay of entropy waves, revealing significant amplitudes at the combustor exit. The presented analysis yields new insights into both generation and transport of entropy waves in partially premixed combustion systems.

CRediT authorship contribution statement

Alexander J. Eder: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Bayu Dharmaputra:** Conceptualization, Validation, Formal analysis, Investigation, Data curation, Writing - review & editing. **Alex M. Garcia:** Conceptualization, Methodology, Software, Writing - review & editing. **Camilo F. Silva:** Conceptualization, Writing - review & editing. **Wolfgang Polifke:** Conceptualization, Validation, Writing - review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We wish to thank CERFACS for providing AVBP, in particular L. Gicquel, O. Vermorel, G. Staffelbach, T. Lesaffre and E. Riber for the technical support. Furthermore, N. Noiray, B. Schuermans and A. Blondé for helpful input on the partially premixed configuration and P. Bonnaire for help with the meshing tool. The authors gratefully acknowledge the German Research Foundation (DFG) for financial support to A. J. Eder under the DFG Transfer Project (No. PO 710/23-1) NoiSI. B. Dharmaputra received funding under the European Research Council (ERC) Grant (No. 820091) TORCH. In addition, the authors gratefully acknowledge the Gauss Centre for Supercomputing e.V. (GCS). for funding this project by providing computing time on the GCS supercomputer SuperMUC-NG at the Leibniz Supercomputing Centre in Garching.

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